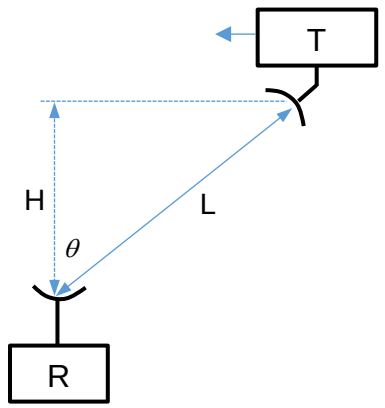
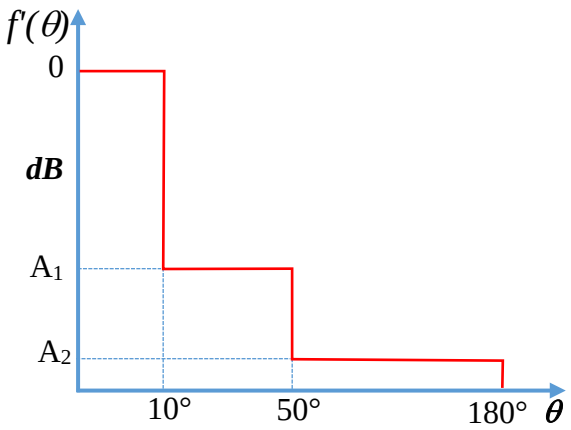
Exercise 3





The transmitter T (P_T=20 W, f_0=4 GHz) is located on an aircraft flying parallel to ground at an altitude H=10 Km. The antenna on T is optimally directed toward the receiving station R on ground and has the gain G_T=10 dB.

The antenna at the ground station R is directed along the vertical direction and exhibits the following directivity diagram: $f(\theta,\varphi) = f'(\theta)\cdot\cos(\varphi/2)$ where $f'(\theta)$ is shown on the figure (A_1= -10 dB). Note that the range of φ is: $-\pi < \varphi < \pi$. The scheme on the above figure (left) refers to the plane φ=0.

1) Compute the parameter A_2 of the directivity diagram in order that the gain of the receiving antenna (G_R) is equal to 17 dB. Assume the radiation impedance Z_R=50 Ω e the loss resistance R_p=10Ω.

3) Compute the power received by R when the aircraft crosses the positions where θ is equal 10° and 50° (with φ=0)
4) It is known the sensitivity of the receiver S=-90 dBm. Evaluate the maximum value of L where the signal is still detected (note that θ>50° for this distance)

Solution:

1) First the directivity gain D_{max} is computed:

$$D_{\max} = \frac{4\pi}{\int_{-\pi}^{\pi}\int_{0}^{\pi}\cos(\varphi/2)\sin(\theta)\,f'(\theta)\,d\theta d\varphi} = \frac{4\pi}{4\left[\left(1-\cos(10°)\right)+A_1\left(\cos(10°)-\cos(50°)\right)+A_2\left(1+\cos(50°)\right)\right]} =$$

$$= \frac{\pi}{0.0494 + A_2 1.6428} = \frac{G}{\eta} = 60.1425$$

Where $\eta=Z_{rad}/(Z_{rad}+R_p)$=0.8333 and G=10^1.7=50.1187
From the above equation it is found A_2=0.00173 (-27.6 dB)

3) At i-th crossing the distance L is equal to:
$L_i=H/\cos(\theta_i)$ with θ_i equal to 10° and 50°. The received power is obtained from the Friis eqution:

$$P_r = P_t + G_T + G_R - L_{f,i} + A_i$$

For θ_i =10°, $L_1=H/\cos(10°)=10.154$ Km, $L_{f,i}=20\log(4\pi L_1/\lambda)=124.61$ dB
For θ_i =50°, $L_2=H/\cos(50°)=15.557$ Km, $L_{f,i}=20\log(4\pi L_2/\lambda)=128.32$ dB

Replacing on the Friis equation (Pt=43dBm, G_T=10dB, G_R=17dB) it has:
Pr1=-64.616 dBm, Pr2=-85.94 dBm

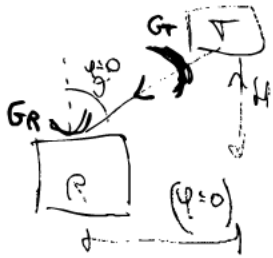
4) From the Friis equation:

$$P_r = P_t + G_T + G_R - L_f + A_2 = -90 \Rightarrow L_f = 132.4 \text{ dB}$$

Then replacing the expression of L_f:

$$L = \frac{\lambda}{4\pi} 10^{L_f/20} = 24.88 \text{ Km}$$

EXERCISE ANTENNAS (N.3)



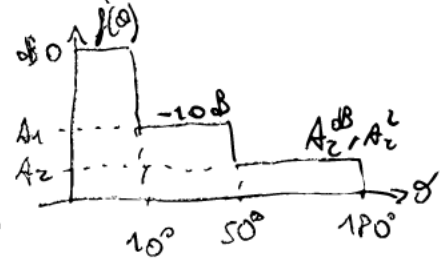
$P_T = 20W$, $f = 4\ GHz$, $H = 10Km$ (AIRCRAFT)

T is OPTIMALLY ALIGNED toward R

$G_T = 10\ dB$

The Receiving ANTENNA

has $f(\vartheta,\varphi) = f'(\vartheta)\cdot\cos\left(\frac{\varphi}{2}\right)$ $\varphi\in[-\pi,\pi]$



① COMPUTE A_2: $G_R = 17\ dB$, with $R_{ADRESISTANCE} = 50\Omega$, $R_{LOSS} = 10\Omega$

$$D = \frac{4\pi r^2}{\oint f(\vartheta,\varphi)\,ds} = \frac{4\pi}{\int_0^{2\pi}\int_0^{\pi} f(\vartheta,\varphi)\,\sin\vartheta\,d\vartheta\,d\varphi}$$

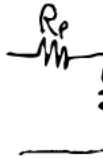
[SEPARABLE]

$$= \frac{4\pi}{\int_{-\pi}^{\pi}\cos\frac{\varphi}{2}\,d\varphi\cdot\int_0^{\pi} f'(\vartheta)\sin\vartheta\,d\vartheta} \qquad \alpha = \frac{\varphi}{2} \Rightarrow d\varphi = 2d\alpha$$

$$= \frac{4\pi}{\int_{-\pi/2}^{\pi/2}(\cos\alpha)2d\alpha\cdot\left(\int_0^{10^\circ}1\sin\vartheta+\int_{10}^{50^\circ}\frac{1}{10}\sin\vartheta+\int_{50}^{180^\circ}\sin\vartheta\,A_2^L\right)}$$

$\varphi = 2\alpha$, $d\varphi = 2d\alpha$

$$= \frac{4\pi}{2\sin\alpha\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cdot\left(-\cos\vartheta\Big|_0^{10^\circ}-\frac{1}{10}\cos\Big|_{10}^{50^\circ}-A_2^L\cos\Big|_{50}^{180^\circ}\right)} \qquad 4\cdot\begin{pmatrix}0.0152\\+0.0342\\+A_2^L\,1.643\end{pmatrix}$$



$P_R = |I|^2\frac{R_R}{2}$, $P_{TOT} = |I|^2\cdot\frac{R_R+R_P}{2} \Rightarrow \eta = \frac{P_R}{P_{TOT}} = \frac{R_R}{R_R+R_P} = 0.83 \Rightarrow$

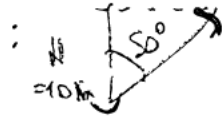
$G = \eta\cdot D = 50.1$

$\Rightarrow A_2^L = 0.0016$

$= -28\ dB$

$D_R = G/\eta = 60.4$

(3) P_{RX} when $\vartheta = 10°$ and $\vartheta = 50°$ (in $\varphi = 0$)

@ 50°: 

$d = \frac{H}{\cos 50°} = 15.55\,Km \rightarrow FSL = 128.3\,dB$

$P_{RX} = P_{TX} + G_{TX} - FSL + 10\log f(0,50°) + G_{RX} = \begin{cases} -68.3\,dBm \\ -86.3\,dBm \end{cases}$ "STEP"

13dBW 10Bi −28÷−10dB 17Bi

@ 10°: $d = \frac{H}{\cos 10°} = 10.2\,Km \rightarrow FSL = 124.6\,dB$

$P_{RX} = \begin{cases} -54.6\,dBm \\ -64.6\,dBm \end{cases}$ "STEP"

(4) Sensitivity is $P_{RX_{MIN}} = -90\,dBm$: what is the MAX DISTANCE? (SLANT DISTANCE)

$\Rightarrow$ it is evidently beyond 50° ($-90\,dBm$)

$P_{TX} + G_{TX} - FSL + G_{RX} + A_z = P_{RX_{MIN}} \Rightarrow FSL = 132.4 \ni = 20\log\frac{4\pi d}{\lambda}$

20dBW = 43dBm 10dB 17dBi −28dB

$d = \frac{\lambda}{4\pi} 10^{\frac{132.4}{20}} \simeq 24.9\,Km$

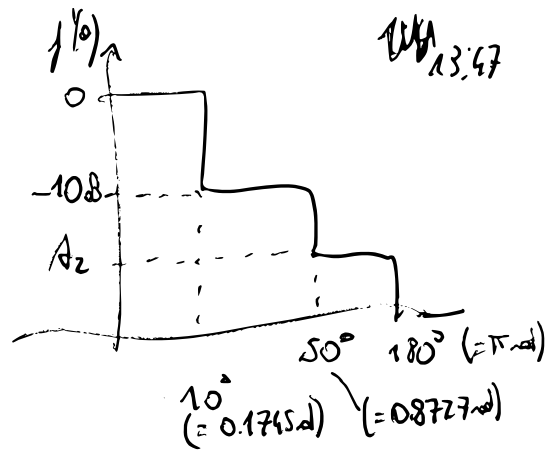
50 minutes

EX 3

$f(\vartheta,\varphi) = f^1(\vartheta)\cdot\cos\left(\frac{\varphi}{2}\right)$

$\varphi \in [-\pi, \pi]$

$\vartheta \in [0, \pi]$



1) Find A_2 so that $G = 17\,dBi$ with $R_{RAD} = 50\Omega$, $R_{LOSS} = 10\Omega$

$$\eta = \frac{R_{RAD}}{R_{RAD}+R_{LOSS}} = 0.833$$



$$\left(P_{RAD} = \frac{V_{RAD}^2}{2R_{RAD}} = \frac{V_{TX}^2 R_{RAD}^2}{2(R_{LOSS}+R_{RAD})^2}\cdot\frac{1}{R_{RAD}} = P_{TX}\frac{R_{RAD}}{R_{RAD}+R_{LOSS}}\right)$$

$$\Rightarrow D_\pi = \frac{G}{\eta} = \frac{10^{\frac{17}{10}}}{\eta} = 60.1 \quad (17.8\,dBi)$$

$$D_M = \frac{4\pi}{\int_0^{2\pi}\int_0^{\pi} f(\vartheta,\varphi)\sin\vartheta\,d\vartheta\,d\varphi} = \frac{4\pi}{\int_{-\pi}^{\pi}\int_0^{\pi} f^1(\vartheta)\cdot\cos\frac{\varphi}{2}\sin\vartheta\,d\vartheta\,d\varphi}$$

$$= \frac{4\pi}{\int_{-\pi}^{\pi}\cos\frac{\varphi}{2}\,d\varphi\cdot\int_0^{\pi} f^1(\vartheta)\sin\vartheta\,d\vartheta} = \frac{4\pi}{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\cos q\,2\,dq\left(\int_0^{0.1745}\sin\vartheta\,d\vartheta + \int_{0.1745}^{0.8727}0.1\sin\vartheta\,d\vartheta + \int_{0.8727}^{\pi}a_2\sin\vartheta\,d\vartheta\right)}$$

$$\frac{\varphi}{2} = q \Rightarrow d\varphi = 2\,dq$$

$$= \frac{4\pi}{2\sin q\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\left(-\cos\vartheta\Big|_0^{0.1745} - 0.1\cos\vartheta\Big|_{0.1745}^{0.8727} - a_2\cos\vartheta\Big|_{0.8727}^{\pi}\right)}$$

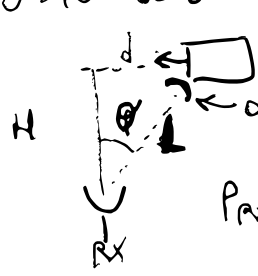
$$= \frac{4\pi}{2\left(1-(-1)\right)\left(1-\cos 0.1745 - 0.1\cos 0.8727 + 0.1\cos 0.1745 - a_2\cos\pi + a_2\cos 0.8727\right)}$$

$$= \frac{\pi}{1-0.9848-0.0643+0.0985+a_2(1+0.6428)}$$

$$= \frac{\pi}{0.0494 + a_2\,1.6428}$$

$$\Rightarrow 60.1 = \frac{\pi}{0.0494 + a_2\cdot 1.6428} \Rightarrow a_2 = \left(\frac{\pi}{60.1} - 0.0494\right)\frac{1}{1.6428} = 0.0017 \quad (= -27.6\,dB)$$

(3) P_{RX} when $\vartheta = 10°$ and $\vartheta = 50°$



$$P_{RX} = P_{TX} + G_T - FSL + G_{RX} + 10\log f(\vartheta,\varphi) \qquad (G_{RX} = 17\,dBi)$$

At 50°: $L = \frac{H}{\cos\vartheta} = 15.6\,km \Rightarrow FSL = 128.3\,dB \Rightarrow P_{RX} = \begin{cases} -115.9\,dBW = -85.9\,dBm \\ -98.3\,dBW = -68.3\,dBm \end{cases}$

At 10°: $L = \frac{H}{\cos\vartheta} = 10.2\,km \Rightarrow FSL = 124.6\,dB \Rightarrow P_{RX} = \begin{cases} -94.6\,dBW = -64.6\,dBm \\ -84.6\,dBW = -54.6\,dBm \end{cases}$

(4) Sensitivity $= -90\,dBm$: Find MAXIMUM L

$\Rightarrow$ We are BEYOND $\vartheta > 50°$! $\div 27.6\,dB$

$S = P_{TX} + G_T - FSL + G + 10\log f(\vartheta,\varphi) \Rightarrow FSL < P_{TX} + G_T + G - 27.6 - S = 132.4\,dB$

($P_{TX} = 43\,dBm$, $S = -90\,dBm$)

$$\Rightarrow FSL = 20\log\frac{4\pi d}{\lambda} \Rightarrow L = \frac{\lambda}{4\pi}\,10^{\frac{132.4}{20}} = 24.9\,km$$